Edge-disjoint barbells from edge-disjoint cycles

Abstract

Let G be a connected multigraph that contains k edge-disjoint cycles. We show that G contains at least $\lceil (k-1)/3 \rceil$ edge-disjoint barbells, where a barbell consists of two edge-disjoint cycles joined by a path. This bound needs no assumption on the lengths of the cycles, and it is exact for infinitely many k. As a corollary, if the cycles have length at most α, with $\alpha \geq 2$ and $k \geq 2$, then there are at least $k/(2\alpha)$ edge-disjoint barbells.

1 Definitions and statement

Graphs are finite multigraphs; parallel edges and self-loops are allowed. A *cycle* of length $L \geq 1$ is a self-loop ($L = 1$), a pair of parallel edges ($L = 2$), or, for $L \geq 3$, a closed path through L distinct vertices. A *path* may consist of a single vertex and no edges. For a subgraph H we write $V(H)$ and $E(H)$ for its vertex and edge sets.

Definition 1 (Barbell)**.** A *barbell* in G is a triple $B = (D_1, D_2, P)$, where D_1, D_2 are edge-disjoint cycles of G and P is a path in G from a vertex of D_1 to a vertex of D_2 such that

(a) $E(P) \cap \big(E(D_1) \cup E(D_2)\big) = \emptyset$, and

(b) no interior vertex of P lies in $V(D_1) \cup V(D_2)$.

The cycles D_1, D_2 are the *ends* of B, and $E(B) := E(D_1) \cup E(D_2) \cup E(P)$. Barbells $B_1, \ldots, B_b$ are *edge-disjoint* if the sets $E(B_i)$ are pairwise disjoint.

If D_1 and D_2 share a vertex, P may be that single vertex (a "figure eight"). Condition (b) is not needed for anything below; we include it only because it makes the conclusion stronger.

Theorem 2. *Let G be a connected multigraph that contains k pairwise edge-disjoint cycles $C_1, \ldots, C_k$. Then G contains at least*

$$\left\lceil \frac{k-1}{3} \right\rceil$$

edge-disjoint barbells, and every end of each of them is one of the cycles C_i. If every C_i has length at most 2, then G contains at least $\lfloor k/2 \rfloor$ edge-disjoint barbells.

Corollary 3. *Let G be a connected multigraph that contains $k \geq 2$ edge-disjoint cycles, each of length at most α, where $\alpha \geq 2$. Then G contains at least $k/(2\alpha)$ edge-disjoint barbells. Without the assumptions $k \geq 2$ and $\alpha \geq 2$, it still contains at least $\lfloor k/(2\alpha) \rfloor$ of them.*

Proof. Let b be the number of barbells given by Theorem 2. If $k \in \{2, 3\}$, then $b \geq \lceil (k-1)/3 \rceil = 1 \geq 3/4 \geq k/(2\alpha)$. If $k \geq 4$, then $b \geq (k-1)/3 \geq k/4 \geq k/(2\alpha)$. For the second claim, if $\alpha \geq 2$ the bound just proved applies whenever $k \geq 2$, and for $k \leq 1$ we have $\lfloor k/(2\alpha) \rfloor = 0$. If $\alpha = 1$, every C_i is a self-loop, and Theorem 2 gives $b \geq \lfloor k/2 \rfloor = \lfloor k/(2\alpha) \rfloor$. □

Remark 4 (The assumptions in Corollary 3 are needed). The real-valued bound $k/(2\alpha)$ can fail if $k = 1$ or $\alpha = 1$. A single triangle has $k = 1$ but no barbell, although $1/(2\alpha) > 0$. A path on three vertices with a self-loop at each vertex has $k = 3$ and $\alpha = 1$, and it has only three cycles in total, so it contains at most one edge-disjoint barbell, which is less than $k/(2\alpha) = 3/2$.

2 Two lemmas

Lemma 5 (From walks to barbells)**.** *Let D_1, D_2 be edge-disjoint cycles of G, and let W be a walk from a vertex of D_1 to a vertex of D_2 with $E(W) \cap (E(D_1) \cup E(D_2)) = \emptyset$. Then there is a path P with $E(P) \subseteq E(W)$ such that (D_1, D_2, P) is a barbell.*

Proof. If D_1 and D_2 share a vertex x, take P to be the path consisting of x alone. Otherwise, every walk contains a path between its endpoints whose edges are edges of the walk, so let $Q = q_0 q_1 \cdots q_s$ be such a path inside W, with $q_0 \in V(D_1)$ and $q_s \in V(D_2)$. Let j be the smallest index with $q_j \in V(D_2)$, and let i be the largest index $i \leq j$ with $q_i \in V(D_1)$; it exists because $q_0 \in V(D_1)$, and $i < j$ because no vertex lies in both cycles. The subpath $P = q_i \cdots q_j$ satisfies condition (a) because $E(P) \subseteq E(W)$, and condition (b) by the choice of i and j. □

Lemma 6 (Disjoint arcs on a cycle)**.** *Let C be a cycle with distinct vertices $v_0, v_1, \ldots, v_{L-1}$ and edges $v_t v_{t+1}$ (indices modulo L), and let $0 < i_1 < i_2 < \cdots < i_r < L$. For $1 \leq j < r$ let A_j be the path $v_{i_j} v_{i_j+1} \cdots v_{i_{j+1}}$, and let A_r be the path $v_{i_r} v_{i_r+1} \cdots v_{L-1} v_0$. Then the arcs $A_1, A_3, A_5, \ldots$ and, if r is odd, A_r, are pairwise edge-disjoint paths in C.*

Proof. The arc A_j uses exactly the edges $v_t v_{t+1}$ with $t \in [i_j, i_{j+1})$, where $i_{r+1} := L$. For the listed indices j these intervals are pairwise disjoint. □

3 Proof of Theorem 2

An auxiliary tree. Call an edge of G a *cycle edge* if it belongs to some C_i, and a *free edge* otherwise. Let A be the graph with vertex set $V(G) \cup \{c_1, \ldots, c_k\}$, where each c_i is a new *cycle node*, and with the following edges:

- every free edge of G, with the same endpoints;
- for every i and every vertex $v \in V(C_i)$, one *star edge* $c_i v$.

The graph A is connected. Indeed, every edge xy of G is either a free edge of A, or an edge of some C_i, and then $x, y \in V(C_i)$ are joined by the path $x\, c_i\, y$ in A. Since G is connected, all of $V(G)$ lies in one component of A, and every c_i is adjacent to a vertex of G.

Fix a spanning tree T of A and root it at a vertex $\rho \in V(G)$. For a node x of T, let T_x be the subtree of its descendants, including x. The neighbors of a cycle node c_i in A are vertices of C_i, and there is only one star edge for each of them. Hence c_i has a parent $p_i \in V(C_i)$, and its children are distinct vertices of C_i different from p_i. The tree edges between two vertices of G are free edges, and a self-loop is never a tree edge.

For a node x, define its *resources* $R_x \subseteq E(G)$ as the set of free edges that are tree edges of T_x, together with all edges of the cycles C_i with $c_i \in T_x$. If x and y are distinct children of the same node, then T_x and T_y are disjoint, and since the cycles C_i are edge-disjoint and free edges lie in no cycle, $R_x \cap R_y = \emptyset$.

Packages. A *package at a vertex* $v \in V(G)$ is a pair (C, W), where C is one of the cycles C_i and W is a walk in G from a vertex of C to v with $E(W) \cap E(C) = \emptyset$. We write $E(C, W) := E(C) \cup E(W)$.

The inductive claim. For a node x of T, let $\hat{x} := x$ if $x \in V(G)$, and $\hat{x} := p_i$ if $x = c_i$. We prove by induction from the leaves of T to the root that every node x admits

- a family $\mathcal{B}_x$ of barbells,
- a set $\mathcal{L}_x$ of cycles (the *lost* cycles), and

- either nothing or one package $\pi_x = (C^x, W^x)$ at $\hat{x}$ (the *exported* package),

with the following properties.

(P1) Every cycle C_i with $c_i \in T_x$ is an end of exactly one barbell of $\mathcal{B}_x$, or lies in $\mathcal{L}_x$, or is C^x; these three cases are mutually exclusive. All ends of barbells in $\mathcal{B}_x$ are cycles C_i with $c_i \in T_x$.

(P2) The sets $E(B)$ for $B \in \mathcal{B}_x$, together with $E(\pi_x)$ if there is an exported package, are pairwise disjoint subsets of R_x.

(P3) $|\mathcal{L}_x| \le |\mathcal{B}_x|$.

Collecting the packages of the children. Let x be a node with children $y_1, \dots, y_s$ (possibly $s = 0$), and assume that the claim holds for every child. We turn the exported packages of the children into packages at a vertex of G:

- If $x \in V(G)$ and $y_j \in V(G)$, the tree edge $e_j = y_j x$ is a free edge. We replace $\pi_{y_j} = (C, W)$, a package at y_j, by (C, We_j), a package at x. Since e_j is free, it is not in $E(C)$.
- If $x \in V(G)$ and $y_j = c_i$ is a cycle node, then $p_i = x$, so π_{y_j} is already a package at x.
- If $x = c_i$ is a cycle node, its child y_j is a vertex of C_i, and π_{y_j} stays a package at y_j.

Call the resulting packages the *arrivals* at x. In each case the edge set of an arrival from y_j is contained in $R_{y_j} \cup \{e_j\}$, and e_j belongs to R_x but to no $R_{y_{j'}}$. Hence, by (P2) for the children, all barbells in $\bigcup_j \mathcal{B}_{y_j}$ and all arrivals have pairwise disjoint edge sets, and these are contained in R_x. Moreover, for an arrival (C, W) from y_j and a cycle C' belonging to an arrival from $y_{j'}$ with $j' \neq j$, we have $E(W) \cap E(C') = \emptyset$.

We start from $\mathcal{B}_x := \bigcup_j \mathcal{B}_{y_j}$ and $\mathcal{L}_x := \bigcup_j \mathcal{L}_{y_j}$, and add to them as follows.

Case 1: $x = v$ is a vertex of G. All arrivals $(C^{(1)}, W^{(1)}), \dots, (C^{(r)}, W^{(r)})$ are packages at v. For $t = 1, 2, \dots, \lfloor r/2 \rfloor$, the walk $W^{(2t-1)}$ followed by the reverse of $W^{(2t)}$ goes from a vertex of $C^{(2t-1)}$ to a vertex of $C^{(2t)}$. It avoids $E(C^{(2t-1)})$ and $E(C^{(2t)})$: each of the two walks avoids its own cycle by the definition of a package, and the other cycle because the two arrivals come from different children. By Lemma 5 we obtain a barbell with ends $C^{(2t-1)}, C^{(2t)}$, whose edges lie in the union of the edge sets of the two arrivals, and we add it to $\mathcal{B}_x$. If r is odd, we export the last arrival, $\pi_x := (C^{(r)}, W^{(r)})$, which is a package at $v = \hat{x}$. Nothing is added to $\mathcal{L}_x$.

Case 2: $x = c_i$ is a cycle node. Write $C := C_i$ and $p := p_i$. The arrivals are packages $(C^{(1)}, W^{(1)}), \dots, (C^{(r)}, W^{(r)})$ at distinct vertices $a_1, \dots, a_r$ of C, all different from p. None of their edges is an edge of C, since they lie in the resources of the children, which do not contain c_i.

- $r = 0$. Export $\pi_x := (C, p)$, where p denotes the walk of length 0 at p. This is a package at $\hat{x} = p$ with edge set $E(C) \subseteq R_x$.
- $r = 1$. The walk $W^{(1)}$ goes from a vertex of $C^{(1)}$ to $a_1 \in V(C)$, and it avoids $E(C^{(1)})$ and $E(C)$. By Lemma 5 we obtain a barbell with ends $C^{(1)}$ and C, whose edges lie in $E(C^{(1)}, W^{(1)}) \cup E(C)$. We add it to $\mathcal{B}_x$ and export nothing.
- $r \ge 2$. Then C contains the $r + 1 \ge 3$ distinct vertices $p, a_1, \dots, a_r$, so its length is $L \ge 3$, and we can write its vertices as $v_0 = p, v_1, \dots, v_{L-1}$ in cyclic order. Relabel the arrivals so that $a_j = v_{i_j}$ with $0 < i_1 < \cdots < i_r < L$, and let $A_1, \dots, A_r$ be the arcs of Lemma 6. For $t = 1, \dots, \lfloor r/2 \rfloor$, the walk

$$W^{(2t-1)}\, A_{2t-1}\, \overline{W^{(2t)}},$$

where the bar denotes reversal, goes from a vertex of $C^{(2t-1)}$ to a vertex of $C^{(2t)}$. Its edges in A_{2t-1} are edges of C, which is edge-disjoint from $C^{(2t-1)}$ and $C^{(2t)}$, and its other edges avoid these two cycles as in Case 1. By Lemma 5 we obtain a barbell with ends $C^{(2t-1)}, C^{(2t)}$, and we add it to $\mathcal{B}_x$. If r is odd, we export $\pi_x := (C^{(r)}, W^{(r)}A_r)$, which is a package at $v_0 = p = \hat{x}$, because A_r ends at v_0 and is edge-disjoint from $C^{(r)}$. Finally we add C to $\mathcal{L}_x$.

Checking the properties. *(P1).* The cycles C_i with $c_i \in T_x$ are the cycles of the subtrees of the children, together with C in Case 2. Each cycle of a child's subtree that was exported by that child is now an end of exactly one new barbell, or is exported by x. The other cycles of the children keep their status. In Case 2 the cycle C is exported ($r = 0$), an end of one new barbell ($r = 1$), or lost ($r \geq 2$).

(P2). Every new barbell and the new exported package use edges of distinct arrivals, together with, in Case 2, edges of C. The arrivals are pairwise edge-disjoint and edge-disjoint from the old barbells. In Case 2 the edges of C belong to R_x but to no R_{y_j}, so they are not used by the arrivals or the old barbells. Within Case 2, they are used only once: by the single barbell when $r = 1$, as the exported cycle when $r = 0$, and, when $r \geq 2$, through the arcs $A_1, A_3, \ldots$ and A_r, which are pairwise edge-disjoint by Lemma 6. All these edges lie in R_x.

(P3). A cycle is added to $\mathcal{L}_x$ only in Case 2 with $r \geq 2$, and then at least $\lfloor r/2 \rfloor \geq 1$ barbells are added to $\mathcal{B}_x$ at the same time.

This completes the induction.

Conclusion. Apply the claim at the root ρ. Since T spans A, every cycle node lies in T_ρ. By (P1), each of the k cycles is an end of exactly one barbell of $\mathcal{B}_\rho$, or lost, or exported by the root. Every barbell has exactly two ends, so with $b := |\mathcal{B}_\rho|$,

$$k \leq 2b + |\mathcal{L}_\rho| + 1 \leq 3b + 1$$

by (P3). Hence $b \geq (k-1)/3$, and since b is an integer, $b \geq \lceil (k-1)/3 \rceil$. By (P2) the barbells are edge-disjoint, and by (P1) their ends are cycles C_i.

If every C_i has length at most 2, then $|V(C_i)| \leq 2$, so Case 2 with $r \geq 2$ never occurs, because it requires three distinct vertices on C. Then $\mathcal{L}_\rho = \emptyset$ and $k \leq 2b + 1$, that is, $b \geq \lfloor k/2 \rfloor$. □

4 The bound is exact

Theorem 7. *For every $m \geq 1$ there is a connected simple graph with $k = 3m$ edge-disjoint triangles that contains at most $m = \lceil (k-1)/3 \rceil$ edge-disjoint barbells.*

Proof. Take a hub vertex h and m gadgets. Gadget g consists of a triangle $Z_g = p_g a_g b_g$, a triangle X_g that meets Z_g only in the vertex a_g, a triangle Y_g that meets Z_g only in b_g, and the edge hp_g. All other vertices are new. The graph G is connected and simple, and every edge lies in at most one cycle. Its cycles are exactly the $3m$ triangles, and they are edge-disjoint.

Consider any family of edge-disjoint barbells. A triangle is an end of at most one of them, because two barbells with a common end share edges. We claim that in each gadget at most two of the three triangles are ends. Suppose that X_g, Y_g and Z_g are all ends. Let B be the barbell with end X_g, let D be its other end, and let P be its path.

In $G - E(Z_g)$, the vertices a_g, b_g and p_g lie in three different components, and $V(X_g)$ lies in the component of a_g. If $D \neq Z_g$, then D is Y_g or a triangle of another gadget, so $V(D)$ lies in the component of b_g or of p_g. The cycles X_g and D have no common vertex, so P joins two different components of $G - E(Z_g)$ and therefore uses an edge of Z_g. But Z_g is an end of a barbell different from B, because B has only the ends X_g and D. This contradicts edge-disjointness, so $D = Z_g$. By the same argument with Y_g in place of X_g, the barbell with end Y_g also has Z_g as its other

end. These two barbells are different, since a barbell has only two ends, and both contain $E(Z_g)$. This is a contradiction.

Hence at most $2m$ triangles are ends, and since every barbell has two ends that are triangles, there are at most m barbells. Finally, $\lceil (3m-1)/3 \rceil = m$, so Theorem 2 is exact for this graph. □

Remark 8. In this example $\alpha = 3$, so the bound $k/(2\alpha) = k/6$ of Corollary 3 is off by a factor of 2, while the bound of Theorem 2 is exact. The length bound α is not needed for Theorem 2 at all; it enters only when the result is compared with $k/(2\alpha)$.